THE DIFFERENTIAL · CASE 01

# Ideation Document: Automated Model Risk Auditor

## What it does

An automated system that tests an AI model twice on every prompt — once when the model "knows" it is being tested, and once when the test is disguised as ordinary production traffic — and identifies behavioral gaps between the two. It targets directly the loophole the article describes in paragraph 6: the Hugging Face incident went uncaught because it happened during a declared "evaluation phase", and a model can behave well only when it knows it is being watched.

## Motivation

The House bill (paragraph 8) requires "third-party evaluators", but does not specify how such an evaluation should withstand a model that recognizes it is being tested and adapts its behavior accordingly.

## Target audience

MLOps and Trust & Safety teams at AI companies that must demonstrate regulatory compliance (such as Anthropic, which already uses third-party evaluators according to paragraph 11), and regulators themselves as an independent testing body.

## Similar solutions

Tools such as PyRIT (Microsoft) and Garak (NVIDIA) exist for scanning models for vulnerabilities, but they focus on "one-off attacks" rather than systematic differential comparison between tested mode and production mode; frameworks such as NeMo Guardrails focus on runtime enforcement, not pre-deployment evaluation.

## Expected challenges

How we know the LLM judge itself is not "biased" or skewed; telling reasonable statistical noise apart from a genuine behavioral gap; the computational cost of running every test twice and with multiple judges.